

```
1 print("-----")
2 print("Element less than 5")
3 x = 0
4 while (x < 5):
5     print(x)
6     x=x+1 x = 5
7
8
9
10 print("-----")
11
12 # Creating a list
13 my_list = [1, 2, 3, "apple", "banana"]
14
15
16 # Modifying elements in a list
17 my_list[2] = 10
18 print("Modified list:", my_list)
19 # Accessing elements in a list
20 print("First element:", my_list[0])
21
22 # Adding elements to a list
23 my_list.append("orange")
24 print("List with added element:", my_list) my_list = [1, 2, 10, 'apple', 'banana', 'orange']
25
26
27
```

```
30
31 num = int(input("Enter a number: "))
32 factorial = 1 factorial = 1
33 for i in range(1, num + 1): num = '6'
34     factorial *= i
35 print("Factorial of", num, "is", factorial)
36
37 print("-----")
38 x = 10
39 if x > 5:
40     print(x,"is greater than 5")
41 elif x == 5:
42     print(x,"is equal to 5")
43 else:
44     print(x," is less than 5")
45
46
47 print("-----")
48
49 # Arithmetic Operations
50 sum = 5 + 3
51 print("Sum:", sum)
52 difference = 10 - 4
53 print("Difference:", difference)
54 product = 6 * 2
55 print("Product:", product)
56 quotient = 12 / 3
57 print("Quotient:", quotient)
58 remainder = 15 % 4
59 print("Remainder:", remainder)
60 power = 2 ** 3
61 print("Power:", power)
62
```

```
PS C:\Users\BMSCECSE\Documents\AI> & 'C:\Users\BMSCECSE\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\BMSCECSE\.vscode\extensions\ms-python.de
bugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '61588' '--' 'C:\Users\BMSCECSE\Documents\AI\1.PY'
Element less than 5
0
1
2
3
4
-----
Modified list: [1, 2, 10, 'apple', 'banana']
First element: 1
List with added element: [1, 2, 10, 'apple', 'banana', 'orange']
-----
Enter a number: 6
Factorial of 6 is 720
-----
10 is greater than 5
-----
Sum: 8
Difference: 6
Product: 12
Quotient: 4.0
Remainder: 3
Power: 8
-----
PS C:\Users\BMSCECSE\Documents\AI>
```